$(\infty, 2)$-CATEGORICAL CONSTRUCTION OF 6-FUNCTOR FORMALISM

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Abstract. These notes are for the talks at AMSS during 2025 spring in the homotopy seminar organized by Shizhang Li, Zhiyou Wu, Weizhe Zheng and Foling Zou. In these talks, we will present the modern ∞-categorical definition of 6-functor formalism using the category of spans. We will recall some basic facts about $(\infty, 2)$-categories and explain how to reform everything into these $(\infty, 2)$-categories. Moreover, we will discuss how to unpack these data into classical 6-functor formalism.

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Introduction

6-functor formalisms are useful tools in algebraic topology and supprisingly help to understand the relationship between geometric objects. It packs geometrical information in an extremely concise fashion, simplifies many of the proofs and enhances cohomology theories as well. Roughly speaking, a classical Grothendick 6-functor formalism should contain the following information.

(1) An underlying category of geometric objects $\mathcal{C}$, with each object X associated with a structure sheaf $\mathrm{D}(X)$.
(2) Each $\mathrm{D}(X)$ admits a closed symmetric monoidal structure $(\mathrm{D}(X), \otimes, \mathcal{I})$, with internal hom $\underline{\mathrm{Hom}}(-,-)$.
(3) For each morphism $f : X \to Y$ in $\mathcal{C}$, there is a **pullback functor** $f^* : \mathrm{D}(Y) \to \mathrm{D}(X)$, which admits a right adjoint $f_* : \mathrm{D}(X) \to \mathrm{D}(Y)$.
(4) For a morphism $f : X \to Y$ in $\mathcal{C}$ satisfying some special conditions, there is a **exceptional pushforward functor** $f_! : \mathrm{D}(X) \to \mathrm{D}(Y)$, which admits a right adjoint $f^! : \mathrm{D}(Y) \to \mathrm{D}(X)$.

These data should satisfy some additional compatibilities. More explicitly, we require

(1) (Symmetric monoidality) f^* commutes with $\otimes$.
(2) (Proper base change) Suppose we have a pullback diagram

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ X' & \xrightarrow[g]{} & X \end{array}$$

Then we have $g^! f_* \cong f'_* g'^!$.
(3) (Projection formula) Given a map $f : Y \to X$ with $M \in \mathrm{D}(X)$ and $N \in \mathrm{D}(Y)$, we have an isomorphism

$$M \otimes f_! N \cong f_!(f^* M \otimes N)$$

Suppose that we already have a 6-functor formalism on an underlying category $\mathcal{C}$, we can define two cohomology theories on $\mathcal{C}$ immediately. One can define the *cohomology* of $X \in \mathcal{C}$ as $\Gamma(X) := p_* \mathbb{1}_X \in \mathrm{D}(*)$ and if X is *smooth*, that means $p\colon X \to *$ admits $p_!$, then the *cohomology with compact support* of X is defined as $\Gamma_c(X) := p_! \mathbb{1}_X$.

Example 0.1. (Künneth formula) Given $X, Y \in \mathcal{C}$ and a 6-functor formalism $\mathrm{D}(-)$ over $\mathcal{C}$, one has

$$\Gamma_c(X \times Y) \simeq \Gamma_c(X) \otimes \Gamma_c(Y)$$

Proof. Consider the following diagram

$$\begin{array}{ccc} X \times Y & \xrightarrow{f_X} & Y \\ {\scriptstyle f_Y}\downarrow & \searrow^{p} & \downarrow{\scriptstyle p_Y} \\ X & \xrightarrow[p_X]{} & * \end{array}$$

Using the proper base change and projection formula, we have

$$\begin{aligned} \Gamma_c(X \times Y) &= p_! \mathbb{1} \simeq p_{Y!} f_{X!} \mathbb{1} \simeq p_{Y!} f_{X!} f^* \mathbb{1} \simeq p_{Y!} p_Y^* p_{X!} \mathbb{1} \\ &\simeq p_{Y!}(p_Y^* p_{X!} \mathbb{1} \otimes \mathbb{1}) \simeq p_{X!} \mathbb{1} \otimes p_{Y!} \mathbb{1} = \Gamma_c(X) \otimes \Gamma_c(Y) \end{aligned}$$

□

Example 0.2. (Poincaré duality) Suppose that X is smooth, then we have

$$p_* p^! \mathbb{1} \simeq \underline{\mathrm{Hom}}(\Gamma_c(X), \mathbb{1})$$

sometimes we refer to $p^! \mathbb{1}$ as ω_X and denote $p_* p^! \mathbb{1}$ by $\Gamma(X, \omega_X)$.

Proof. Using the base-change and projection formula, we have

$$\mathrm{Map}(M, p_* p^! \mathbb{1}) \simeq \mathrm{Map}(p_! p^* M, \mathbb{1}) \simeq \mathrm{Map}(M \otimes p_! \mathbb{1}, \mathbb{1}) \simeq \mathrm{Map}(M, \underline{\mathrm{Hom}}(\Gamma_c(X), \mathbb{1}))$$

Then we conclude by Yoneda. □

Following [HM24], the contents are divided as follows. In Section 1 we define the basic notations and introduce the ∞-categorical definition of a 6-functor formalism. We start with the main ingredient: the category of spans(or correspondences) in Section 1.1. Then we define a 6-functor formalism as a map of operads. We explain how to upgrade a 6-functor

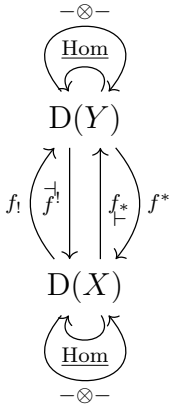


FIGURE 1. A six-functor formalism

formalism to a 2-category in Section 2 and explain the associated properties. In Section 3 we explain the descent properties.

1. 6-functor fomalisms

In this section, we explain the definition of 6-functor formalisms. Informally, a 6-functor formalism can be thought as a Cat_∞-valued functor with double value on morphisms. we will make this idea precise using the language of category of spans.

1.1. Category of spans.

The term "spans" is usually used to denote diagrams of the form $\bullet \leftarrow \bullet \rightarrow \bullet$. However, in many circumstances we will need a more general form of spans, where two arrows are weighted differently. To solve this, we will need the following definition.

Definition 1.1. Let $\mathcal{C}$ be am ∞-category and $\mathcal{C}_0$ be a wide subcategory, i.e. a subcategory with all objects of $\mathcal{C}$. We say that the pair $(\mathcal{C},\mathcal{C}_0)$ is a *geometric setup* if

(1) $\mathcal{C}_0$ is closed under pullbacks along morphisms in $\mathcal{C}$.
(2) The inclusion $\mathcal{C}_0 \to \mathcal{C}$ preserves pullbacks.

Trivial examples of geometric setups include $(\mathcal{C},\mathcal{C}^{\simeq})$ where $\mathcal{C}^{\simeq}$ denotes the underlying anima of $\mathcal{C}$ and $(\mathcal{C},\mathcal{C})$. There are many nontrivial examples in algebraic geometry because one can consider those $\mathcal{C}_0$ living with specific morphisms that are stable under base change.

Geometry setups are organized into a category Gst, which becomes a subcategory of $\mathrm{Fun}([1],\mathrm{Cat})$ through obvious inclusion. The category Gst is complete. The limits can be computed termwise and the inclusion $\mathrm{Gst} \to \mathrm{Fun}([1],\mathrm{Cat})$ preserves limits.

In order to form the infinity categorical definition of spans, we will need the following notation

Definition 1.2. We have the following posets:

(1) We denote by $\mathbb{\Sigma}^n$ the poset of pairs (i,j) with $0 \leqslant i \leqslant j \leqslant n$. The partial order is given by $(i,j) \leqslant (k,l)$ if $i \leqslant k$ and $j \leqslant l$. $\mathbb{\Sigma}^\bullet$ becomes a cosimplicial object in Cat.
(2) We denote by$\mathbb{\Sigma}^n_{\simeq} \subset \mathbb{\Sigma}^n$ the subposet where the partial order is given by $(i,j) \leqslant (k,l)$ if $i \leqslant k$ and $j = l$. $\mathbb{\Sigma}^\bullet_{\simeq} \subset \mathbb{\Sigma}^\bullet$ becomes a cosimplicial object.
(3) We denote by $\mathbb{\Lambda}^n \subset \mathbb{\Sigma}^n$ the full subcategory consisting of pairs (i,j) with $j - i \leqslant 1$.

Note that $(\mathbb{\Sigma}^n, \mathbb{\Sigma}^n_{\simeq})$ is a geometric setup for $n \geqslant 0$ as in Definition 1.1. Therefore, one obtains a cosimplicial object $(\mathbb{\Sigma}^\bullet, \mathbb{\Sigma}^\bullet_{\simeq})$ in Gst.

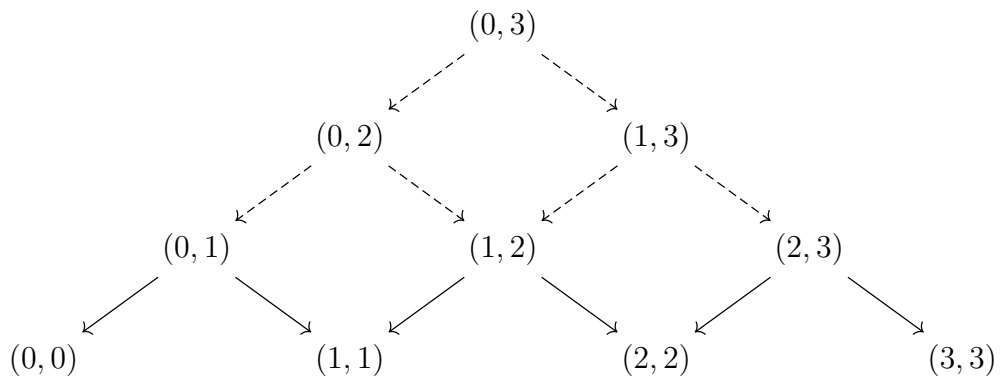


FIGURE 2. The diagrams depicts $\mathbb{\Sigma}^3$ (all arrows) and $\mathbb{\Lambda}^3$ (solid arrows). The poset $\mathbb{\Sigma}^3_{\simeq}$ contains only arrows pointing to the right.

Definition 1.3. The composite

$$\Delta^{\mathrm{op}} \times \mathrm{Gst} \xrightarrow{(\mathbb{\Sigma}^\bullet, \mathbb{\Sigma}^\bullet_{\simeq})\times \mathrm{id}} \mathrm{Gst}^{\mathrm{op}} \times \mathrm{Gst} \xrightarrow{\mathrm{Map}(-,-)} \mathcal{S}$$

defines a functor $\mathrm{Span} : \mathrm{Gst} \to \mathrm{Fun}(\Delta^{\mathrm{op}}, \mathcal{S})$, which we refer to as the *span functor*.

For any geometric setup $(\mathcal{C}, \mathcal{C}_0)$, $\mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ takes value in Cat. To see this, let us review the notation of complete Segal anima.

Definition 1.4. A simplicial anima $A_\bullet : \Delta^{\mathrm{op}} \to \mathcal{S}$ is called a *Segal anima* if it satisfies the following Segal condition:

$$A_n \xrightarrow[\simeq]{(\rho_1, \cdots, \rho_n)} A_1 \times_{A_0} \cdots \times_{A_0} A_1$$

where ρ_i is induced by the inclusion $\{i-1, i\} \to [n]$. Let J be a diagram with two points and a unique isomorphism between them. We say a Segal anima is complete if $N(J) \to \Delta^0$ induces an isomorphism

$$A_0 \simeq \mathrm{Map}(\Delta^0, A^\bullet) \simeq \mathrm{Map}(N(J), A^\bullet)$$

Theorem 1.5 ([JT07];Theorem 4.12)**.** *The full subcategory of* $\mathrm{Fun}(\Delta^0, \mathcal{S})$ *consisting of complete Segal animas is identified with* Cat *under the Rezk nerve.*

We can show that the simplicial anima $\mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ is a complete Segal anima. The "Segal" part follows from the observation that we have an isomorphism of spaces

$$\mathrm{Map}_{\mathrm{Gst}}((\mathbb{\Sigma}^n, \mathbb{\Sigma}^n_{\simeq}), (\mathcal{C}, \mathcal{C}_0)) \simeq \mathrm{Fun}^{\mathrm{Cart}, \mathcal{C}_0}(\mathbb{\Sigma}_n, \mathcal{C})^{\simeq} \simeq \mathrm{Fun}^{\mathcal{C}_0}(\mathbb{\Lambda}_n, \mathcal{C})^{\simeq}$$

where $\mathrm{Fun}^{\mathcal{C}_0}(\mathbb{\Lambda}_n, \mathcal{C})$ denotes the functors that send right arrows into $\mathcal{C}_0$ and $\mathrm{Fun}^{\mathrm{Cart}, \mathcal{C}_0}(\mathbb{\Sigma}_n, \mathcal{C})^{\simeq}$ denotes additionally the functors that send the smallest squares to cartesian squares. Therefore, to show that $\mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ is a Segal anima, it suffices to show that we have an equivalence

$$\mathrm{Map}(\Delta^n, \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)) \simeq \mathrm{Map}(I^n, \mathrm{Span}(\mathcal{C}, \mathcal{C}_0))$$

By definition, the left side is identified with $\mathrm{Fun}^{\mathrm{Cart}, \mathcal{C}_0}(\mathbb{\Sigma}_n, \mathcal{C})^{\simeq}$ and the right side is identified with $\mathrm{Fun}^{\mathcal{C}_0}(\mathbb{\Lambda}_1, \mathcal{C})^{\simeq} \times_{\mathcal{C}} \cdots \times_{\mathcal{C}} \mathrm{Fun}^{\mathcal{C}_0}(\mathbb{\Lambda}_1, \mathcal{C})^{\simeq}$s, which naturally equivalent to $\mathrm{Fun}^{\mathcal{C}_0}(\mathbb{\Lambda}_n, \mathcal{C})^{\simeq}$ since $\mathbb{\Lambda}_n$ is formed by piecing n copies of $\mathbb{\Lambda}_1$ together. Then both sides are identified. The "complete" part follows from [Hau18, Proposition 8.1].

Now we discuss some properties of the category of spans. One can first right down the explicit description for $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$. The objects of $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$ are the same as that of $\mathcal{C}$, while the n-cells of $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$ are diagrams $\mathrm{Fun}^{\mathrm{Cart},\mathcal{C}_0}(\mathbb{\Sigma}_n,\mathcal{C})^{\simeq}$.

By definition, we have

Lemma 1.6. *The functor* $\mathrm{Span}\colon \mathrm{Gst}\to\mathrm{Cat}$ *preserves limits.*

Example 1.7. We have following immediate examples:

(1) By precomposing with $(\mathbb{\Sigma}^\bullet,\mathbb{\Sigma}^\bullet_{\simeq})\to([-],[-])$, one obtains an embedding
$$\mathcal{C}_0\to\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$$
which sends $X\to Y$ to $X\xleftarrow{\mathrm{id}}X\to Y$.

(2) Similarly, we have an embedding
$$\mathcal{C}^{\mathrm{op}}\to\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$$
which sends $X\to Y$ to $Y\leftarrow X\xrightarrow{\mathrm{id}}X$.

(3) The non-trivial isomorphism $\mathbb{\Sigma}^{\bullet,\mathrm{op}}\simeq\mathbb{\Sigma}^\bullet$ induces an equivalence
$$\mathrm{Span}(\mathcal{C},\mathcal{C}_0)\simeq\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\mathrm{op}}$$

We can assign a highly none-trivial operad structure over $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$, which carries a lot of information and allow us to discuss the 2-categorical structures. Let $(\mathcal{C},\mathcal{C}_0)\in\mathrm{Gst}$, we consider the coCartesian operad$(\mathcal{C}^{\mathrm{op}})^{\sqcup}\to\mathrm{Fin}_*$ and denote the maps in $(\mathcal{C}^{\mathrm{op}})^{\sqcup,\mathrm{op}}_{\langle n\rangle}$ by $f: X_1\oplus\cdots\oplus X_n\to Y_1\oplus\cdots\oplus Y_n$.

Notation 1.8. Let $(\mathcal{C},\mathcal{C}_0)$ be a geometric setup. We put
$$\mathcal{C}_0^-:=(\mathcal{C}_0^{\mathrm{op}})^{\sqcup,\mathrm{op}}\times_{\mathrm{Fin}_*^{\mathrm{op}}}\mathrm{Fin}_*^{\simeq}$$
It follows from the definition and the construction that $((\mathcal{C}^{\mathrm{op}})^{\sqcup,\mathrm{op}},\mathcal{C}_0^-)$ is also a geometric setup.

Definition 1.9. Let $(\mathcal{C},\mathcal{C}_0)$ be a geometric setup. We put
$$\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}:=\mathrm{Span}((\mathcal{C}^{\mathrm{op}})^{\sqcup,\mathrm{op}},\mathcal{C}_0^-)$$
If we translate this definition, we find that the objects of $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$ are tuples of the form $X_\bullet=(X_1,\cdots,X_n)$ of objects in $\mathcal{C}$ and morphisms from $X_\bullet$ to $Z_\bullet$ are spans $X_\bullet\leftarrow Y_\bullet\to Z_\bullet$ with $Y_\bullet$ and $X_\bullet$ being of the same size.

Proposition 1.10. *Let* $(\mathcal{C},\mathcal{C}_0)$ *be a geometric setup. Then* $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$ *becomes an operad with the underlying category identified with* $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$*. Moreover, if* $\mathcal{C}$ *admits finite products, then* $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$ *becomes a symmetric monoidal category.*

Proof. We first show that $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$ is an operad over $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$. We need to verify the operad axioms. Denote by $p\colon\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to\mathrm{Fin}_*$ the obvious projection. First, for any inert edge $f\colon\langle n\rangle\to\langle m\rangle$ in Fin_*, we can find some p-coCartesian edge $\overline{f}\in(\mathcal{C}^{\mathrm{op}})^{\sqcup}$ lifting f, which implies that $\bullet\xleftarrow{\overline{f}}\bullet\xrightarrow{\mathrm{id}}\bullet$ is a p-coCartesian lifting of f in $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$. Secondly, given $X_1\cdots,X_n\in\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$, we can find $X\in(\mathcal{C}^{\mathrm{op}})^{\sqcup}$ such that $X\to X_i$ is a p-coCartesian lifting of $\rho_i\colon\langle n\rangle\to\langle 1\rangle$, which also gives a p-coCartesian lifting of ρ_i in $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$. Finally,

for any $f\colon \langle n\rangle \to \langle m\rangle$ in Fin_* and $X, Y \in \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)^{\otimes}$ living over $\langle n\rangle$ and $\langle m\rangle$ respectively, we need to show that the map

$$\mathrm{Map}^f(X,Y) \to \prod_{i=1}^{m} \mathrm{Map}^{\rho_i \circ f}(X, Y_i)$$

induced by ρ_i is an equivalence. This follows from

$$\begin{aligned}
\mathrm{Map}^f(X,Y) &\simeq \left((\mathcal{C}^{\mathrm{op}})^{\sqcup}_{X/f} \times_{(\mathcal{C}^{\mathrm{op}})^{\sqcup}} (\mathcal{C}_0^{-})^{\mathrm{op}}_{Y/\mathrm{id}} \right)^{\simeq} \\
&\simeq \left((\mathcal{C}^{\mathrm{op}})^{\sqcup}_{\langle m\rangle, X/f} \times_{(\mathcal{C}^{\mathrm{op}})^{\sqcup}_{\langle m\rangle}} \prod_{i=1}^{m} (\mathcal{C}_0)^{\mathrm{op}}_{Y_i/} \right)^{\simeq} \\
&\simeq \left(\prod_{i=1}^{m} (\mathcal{C}^{\mathrm{op}})^{\sqcup}_{\langle m\rangle, X/\rho_i\circ f} \times_{(\mathcal{C}^{\mathrm{op}})^{\sqcup}} (\mathcal{C}_0)^{\mathrm{op}}_{Y_i/} \right)^{\simeq} \\
&\simeq \mathrm{Map}^{\rho_i \circ f}(X, Y_i)
\end{aligned}$$

where $(\mathcal{C}^{\mathrm{op}})^{\sqcup}_{-/f}$ denotes the full subcategory of $(\mathcal{C}^{\mathrm{op}})^{\sqcup}_{-/}$ consisting of maps that lift f.

The symmetric monoidal part is similar from the first argument. □

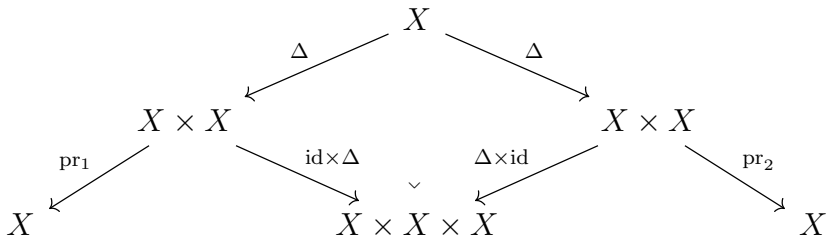
Remark 1.11. The symmetric monoidal structure given by $\mathrm{Span}(\mathcal{C}, \mathcal{C}_0)^{\otimes}$ is neither Cartesian or coCartesian. We will soon see that this carries an extraordinary 2-categorical structure of a 6-functor formalism.

The assignment $(\mathcal{C}, \mathcal{C}_0) \mapsto \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)^{\otimes}$ upgrades to a functor $\mathrm{Gst} \to \mathrm{Op}$. Moreover, the embeddings $\mathcal{C}_0 \to \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ and $\mathcal{C}^{\mathrm{op}} \to \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ upgrade to operad maps $\mathcal{C}_0^{\times} \to \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)^{\otimes}$ and $(\mathcal{C}^{\mathrm{op}})^{\sqcup} \to \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)^{\otimes}$.

From now on we suppose that $\mathcal{C}$ admits finite products. Therefore, all operad maps become symmetric monoidal functors. We discuss the self-enrichment property under this hypothesis.

Proposition 1.12. *Let $(\mathcal{C}, \mathcal{C}_0)$ be a geometric setup such that $X \to *$ lies in $\mathcal{C}_0$, where $*$ denotes the final object. Then X is dualizable in $\mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ with $X^{\vee} = X$.*

Proof. The condition that $X \to *$ lies in $\mathcal{C}_0$ also guarantees that the diagonal map$\Delta\colon X \to X \times X$ lies in $\mathcal{C}_0$, then we can take $ev_X = [X \times X \xleftarrow{\Delta} X \to *]$ and $coev_X = [* \leftarrow X \xrightarrow{\Delta} X \times X]$. It remains to show the unit law and the counit law. We prove the former since the latter is similar. By unpacking the definition, it suffices to show that the middle square of



is a pullback square, which is clear by simple diagram trace. □

Corollary 1.13. *Let $\mathrm{Span}(\mathcal{C}) := \mathrm{Span}(\mathcal{C}, \mathcal{C})$ and let $\mathcal{C}$ admit finite limits. Then $\mathrm{Span}(\mathcal{C})^{\otimes}$ is closed symmetric monoidal with $\underline{\mathrm{Hom}}(X, Y) = Y \times X$.*

Proof. This is because in this case every object is self-dual. □

Corollary 1.14. *Under the above assumption, $\mathrm{Span}(\mathcal{C})$ is enriched over itself.*

1.2. **3-functor formalisms.** We now discuss how to define a 6-functor formalism using category of spans. Recall that a 6-functor formalism should contain the following data:

(1) An underlying category of geometric objects $\mathcal{C}$, with each object X associated with a structure sheaf $\mathrm{D}(X)$ where each $\mathrm{D}(X)$ admits a closed symmetric monoidal structure $(\mathrm{D}(X),\otimes,\mathcal{I})$, with internal hom $\underline{\mathrm{Hom}}(-,-)$.
(2) For each morphism $f: X\to Y$ in $\mathcal{C}$, there is a functor $f^*:\mathrm{D}(Y)\to\mathrm{D}(X)$, which admits a right adjoint $f_*:\mathrm{D}(X)\to\mathrm{D}(Y)$.
(3) For a morphism $f: X\to Y$ in $\mathcal{C}$ satisfying some special conditions, there is a functor $f_!:\mathrm{D}(X)\to\mathrm{D}(Y)$, which admits a right adjoint $f^!:\mathrm{D}(Y)\to\mathrm{D}(X)$.

These data should satisfy some compatibilities and functorialities, which inspires the following concise definition:

Definition 1.15. Let $(\mathcal{C},\mathcal{C}_0)$ be a geometric setup. A *3-functor formalism* on $(\mathcal{C},\mathcal{C}_0)$ is a map of operad

$$\mathrm{D}(-):\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to\mathrm{Cat}^{\times}$$

where $\mathrm{Cat}^{\times}$ denotes the Cartesian operad on the category of categories.

Remark 1.16. When $\mathcal{C}$ admits finite limits, then $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$ becomes a symmetric monoidal category and hence a 3-functor formalism is a lax symmetric monoidal functor $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)\to\mathrm{Cat}^{\times}$.

We can read out the desired data from this definition. More precisely, suppose we have an operad map $\mathrm{D}(-):\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to\mathrm{Cat}^{\times}$, precomposing with $\mathcal{C}^{\mathrm{op},\sqcup}\to\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$ gives an operad map $\mathcal{C}^{\mathrm{op},\sqcup}\to\mathrm{Cat}^{\times}$. By [Lur17, Theorem 2.4.3.18], this is equivalent to a functor $\mathcal{C}^{\mathrm{op}}\to\mathrm{CMon}$, which gives the desired symmetric monoidal category $(\mathrm{D}(X),\otimes,\mathcal{I})$ for each $X\in\mathcal{C}$.

For each $f: X\to Y\in\mathcal{C}$, the precomposition $\mathrm{D}(-):\mathcal{C}^{\mathrm{op},\sqcup}\to\mathrm{CMon}$ gives the desired pullback $f^*:\mathrm{D}(Y)\to\mathrm{D}(X)$. Similarly, precomposing with $\mathcal{C}_0\to\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$ gives the desired exceptional pushforward $f_!$.

By considering the suitable diagram and unpacking the definition, one can also spell out the desired properties from the definition.

Proposition 1.17. *Let $\mathrm{D}(-):\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to\mathrm{Cat}^{\times}$ be a 3-functor formalism over $(\mathcal{C},\mathcal{C}_0)$, then $\otimes$, f^* and $f_!$ satisfies the following:*

(1) (Functoriality) For f,g such that the notation makes sense, we have $(f\circ g)^*\simeq g^*\circ f^*$ and $(f\circ g)_!\simeq f_!\circ g_!$.
(2) (Symmetric monoidal) f^* is symmetric monoidal with respect to $\otimes$.
(3) (Proper base change) For every Cartesian square

$$\begin{array}{ccc} A & \xrightarrow{g'} & B \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ C & \xrightarrow[g]{} & D \end{array}$$

with $f\in\mathcal{C}_0$, we have a natural equivalence

$$g^*f_!\simeq f'_!g'^*$$

(4) (Projection formula) For f that makes sense, we have

$$M\otimes f_!N\simeq f_!(f^*M\otimes N)$$

1.3. **6-functor formalisms.** A 6-functor formalism is a 3-functor formalism with additional structure consisting of $\underline{\mathrm{Hom}}(-,-)$, f_* and $f^!$. More formally, we have

Definition 1.18. Let $(\mathcal{C},\mathcal{C}_0)\in \mathrm{Gst}$. A 6-functor formalism on $(\mathcal{C},\mathcal{C}_0)$ is a 3-functor formalism $\mathrm{D}(-)\colon \mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to \mathrm{Cat}^{\times}$ with the following structures:

(1) For every $X\in\mathcal{C}$, $\mathrm{D}(X)$ is closed symmetric monoidal. We denote the right adjoint of $-\otimes M$ by $\underline{\mathrm{Hom}}(M,-)$.
(2) For each $f\in\mathcal{C}$, f^* admits a right adjoint f_*.
(3) For each $g\in\mathcal{C}_0$, $g_!$ admits a left adjoint $f^!$.

Remark 1.19. Suppose that a 3-functor formalism $\mathrm{D}(-)\colon \mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to \mathrm{Cat}^{\times}$ factors through $(\mathrm{Pr}^L)^{\otimes}$, the symmetric monoidal category of presentable categories and left adjoints equipped with Lurie's tensor. Then $\mathrm{D}(-)$ naturally acquires a structure of 6-functor formalism.

A 6-functor formalism will satisfy the following properties

Proposition 1.20. *Let $\mathrm{D}(-)\colon \mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to \mathrm{Cat}^{\times}$ be a 6-functor formalism over $(\mathcal{C},\mathcal{C}_0)$, then we have the following properties:*

(1) (Functoriality) For f,g such that the notation makes sense, we have $(f\circ g)_*\simeq f_*\circ g_*$ and $(f\circ g)^!\simeq g^!\circ f^!$.
(2) (Smooth base change) For every Cartesian square

$$\begin{array}{ccc} A & \xrightarrow{g'} & B \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ C & \xrightarrow[g]{} & D \end{array}$$

with $g\in\mathcal{C}_0$, we have a natural equivalence

$$g^! f_*\simeq f'_* g'^!$$

(3) For f that makes sense, we have

$$\underline{\mathrm{Hom}}(f_! N,M)\simeq f_*\underline{\mathrm{Hom}}(N,f^! M)$$

(4) For f that makes sense, we have

$$f^!\underline{\mathrm{Hom}}(N,M)\simeq \underline{\mathrm{Hom}}(f^* N,f^! M)$$

Proof. The claim (1) and (2) follows from passing Proposition 1.17 to right adjoint. We prove the claim (3). For any $P\in\mathrm{D}(X)$, by proper base change, we have

$$\begin{aligned}\mathrm{Map}(P,\underline{\mathrm{Hom}}(f_! N,M))\simeq&\mathrm{Map}(P\otimes f_! N,M)\simeq \mathrm{Map}(f_!(f^* P\otimes N),M)\simeq \mathrm{Map}(f^* P\otimes N,f^! M)\\ \simeq&\mathrm{Map}(f^* P,\underline{\mathrm{Hom}}(N,f^! M))\simeq \mathrm{Map}(P,f_*\underline{\mathrm{Hom}}(N,f^! M))\end{aligned}$$

so we conclude by Yoneda. The last claim is similar. □

2. Upgrade to $(\infty,2)$-category

We now discuss the 2-category structure associated with a 6-functor formalism. Let $\mathcal{C}$ be a category that admits finite limits. For a 6-functor formalism $\mathrm{D}(-)$ over $(\mathcal{C},\mathcal{C})$, there is a 2-category attached to $\mathrm{D}(-)$, which we refer to as the category of kernels. It first emerges in [LZ22] and surprisingly contains a lot of geometrical data.

2.1. **Recollection of $(\infty,2)$-category.** We first recall some facts about the $(\infty,2)$-category. We begin with some recollections of ∞-categorical version of categorical enrichment, where we adopt Lurie's model.

We denote by $\mathrm{LM}^{\otimes}$ the left module operad defined in [Lur17, Notation 4.2.1.6], let $\langle \mathfrak{a}, \mathfrak{m} \rangle$ denote the two color of it and let $\mathrm{Ass}^{\otimes} \to \mathrm{LM}^{\otimes}$ be the map over the first color.

Definition 2.1. Let $\mathcal{O}^{\otimes} \to \mathrm{LM}^{\otimes}$ be a map of operads and let $\mathcal{V}^{\otimes} := \mathcal{O}^{\otimes} \times_{\mathrm{LM}^{\otimes}} \mathrm{Ass}^{\otimes}$. We also denote by $\mathcal{C} := \mathcal{O}_{\mathfrak{m}}$ the underlying category.

(1) A *morphism object* of $X, Y \in \mathcal{C}$ in $\mathcal{O}^{\otimes}$ is an object $\mathrm{Hom}^{\mathcal{V}}(X,Y) \in \mathcal{V}$ together with a map in $\mathrm{Mul}_{\mathcal{V}}(\mathrm{Hom}^{\mathcal{V}}(X,Y), X; Y)$ which induces an equivalence
$$\mathrm{Mul}_{\mathcal{V}}(V_1, \cdots, V_n; \mathrm{Hom}^{\mathcal{V}}(X,Y)) \xrightarrow{\simeq} \mathrm{Mul}_{\mathcal{V}}(V_1, \cdots, V_n, X; Y)$$
for all $V_1, \cdots, V_n \in \mathcal{V}$.

(2) We say that $\mathcal{O}^{\otimes}$ exhibits $\mathcal{C}$ as *enriched* over $\mathcal{V}$ if for every $X, Y \in \mathcal{C}$, there is a morphism object $\mathrm{Hom}^{\mathcal{V}}(X,Y)$.

We denote by $\mathrm{Enr} \subset \mathrm{Op}_{/\mathrm{LM}^{\otimes}}$ the full subcategory spanned by the operads $\mathcal{O}^{\otimes}$ that exhibits $\mathcal{C} := \mathcal{O}_{\mathfrak{m}}$ as enriched over $\mathcal{V}^{\otimes} = \mathcal{O}^{\otimes} \times_{\mathrm{LM}^{\otimes}} \mathrm{Ass}^{\otimes}$. We denote by $\mathrm{Enr}_{\mathcal{V}} \subset \mathrm{Enr}$ the fiber of $\mathcal{V}$ along the forgetful functor $\mathrm{Enr} \to \mathrm{Op}_{/\mathrm{LM}^{\otimes}}$ and call it the category of $\mathcal{V}$-*enriched category*.

Remark 2.2. There are different models for categorical enrichment. In [GH15] and [Hin18], the authors developed another approach to the enrich category using the construction of quivers. The two methods are equivalent as shown in [Hei23]. It is convenient for us to choose any model we like.

Definition 2.3. In the setting of Definition 2.1, let $\mathcal{C} = \mathcal{O}^{\otimes} \to \mathrm{LM}^{\otimes}$ be a $\mathcal{V}$-enriched category. We call $\underline{\mathcal{C}} := \mathcal{O}^{\otimes}_{\mathfrak{m}}$ the underlying category of $\mathcal{O}^{\otimes}$. This construction provides a functor $\mathrm{Enr} \to \mathrm{Cat}$.

Example 2.4. In classical category theory, all categories are enriched over Set. The analogy for ∞-category also holds true. That is, we have
$$\mathrm{Enr}_{\mathcal{S}} \simeq \mathrm{Cat}$$
as shown in [Hei23].

Definition 2.5. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be two $\mathcal{V}$-enriched categories and let $\mathcal{O}_1^{\otimes}, \mathcal{O}_2^{\otimes} \to \mathrm{LM}^{\otimes}$ be the associated operads. We denote by
$$\mathrm{Fun}^{\mathcal{V}}(\mathcal{C}_1, \mathcal{C}_2) := \mathrm{Alg}_{\mathrm{LM}^{\otimes}}(\mathcal{O}_1^{\otimes}, \mathcal{O}_2^{\otimes}) \times_{\mathrm{Alg}_{\mathrm{Ass}^{\otimes}}(\mathcal{V},\mathcal{V})} \{\mathrm{id}\}$$
and call it the category of $\mathcal{V}$-*enriched functors* from $\mathcal{C}_1$ to $\mathcal{C}_2$.

By passing to the underlying category, we can also talk about the underlying functor between two underlying categories of a $\mathcal{V}$-enriched functor.

The composition $\mathrm{Enr} \to \mathrm{Op}_{/\mathrm{LM}^{\otimes}} \to$ is in fact a coCartesian fibration. Using this, we can easily define transfer of enrichment.

Definition 2.6. Let $\alpha \colon \mathcal{V}^{\otimes} \to \mathcal{W}^{\otimes}$ be a map of associative operads. Straightening along the coCartesian fibration $\mathrm{Enr} \to \mathrm{Op}_{/\mathrm{Ass}^{\otimes}}$, we obtain a functor
$$\tau_{\alpha} \colon \mathrm{Enr}_{\mathcal{V}} \to \mathrm{Enr}_{\mathcal{W}}$$
and call it the *transfer of enrichment* along α.

Unpacking the definition, one can find that for any $\mathcal{V}$-enriched category $\mathcal{C}$, the $\mathcal{W}$-enriched category $\tau_\alpha(\mathcal{C})$ has the same object as $\mathcal{C}$ with the $\mathcal{V}$-enriched morphism objects given by

$$\mathrm{Hom}^{\mathcal{W}}_{\tau_\alpha(\mathcal{C})}(X,Y) \simeq \alpha\left(\mathrm{Hom}^{\mathcal{V}}_{\mathcal{C}}(X,Y)\right)$$

An interesting property of transfer of enrichment is that it can decompose maps of operads. We now make it more precise.

Lemma 2.7. *Let $\alpha\colon \mathcal{V} \to \mathcal{W}$ be a lax monoidal functor between two monoidal categories.*

(1) For every $\mathcal{V}$-enriched category $\mathcal{C}$, applying Enr $\to$ Cat *to the edge $\mathcal{C} \to \tau_\alpha(\mathcal{C})$, we obtain an essentially surjective functor*

$$F_\alpha\colon \underline{\mathcal{C}} \to \underline{\tau_\alpha(\mathcal{C})}$$

(2) The natural assignment $X \to \alpha(X)$ upgrades to a $\mathcal{W}$-enriched functor

$$G_\alpha\colon \tau_\alpha(\mathcal{V}) \to \mathcal{W}$$

(3) Viewing $\mathcal{V}$ as a self-enriched category, then α can be decomposed as

$$\mathcal{V} = \underline{\mathcal{V}} \xrightarrow{F_\alpha} \underline{\tau_\alpha(\mathcal{V})} \xrightarrow{G_\alpha} \underline{\mathcal{W}} = \mathcal{W}$$

Proof idea. One can take $G_\alpha = \mathrm{Hom}^{\mathcal{W}}_{\tau_\alpha(\mathcal{W})}(\mathbb{1}_{\mathcal{V}}, -)$ and check the properties one by one. □

We are now prepared to define the language of $(\infty, 2)$-categories.

Definition 2.8. We use Lurie's model of enriched categories.

(1) A 2-category is a Cat-enriched category and a 2 -functor between 2-categories is a Cat-enriched functor. We denote

$$\mathrm{Cat}_2 := \mathrm{Enr}_{\mathrm{Cat}}$$

(2) For any 2-category $\mathcal{C} \in \mathrm{Cat}_2$ and any $X, Y \in \mathcal{C}$ we define the morphism category from X to Y to be

$$\mathrm{Fun}_{\mathcal{C}}(X,Y) := \mathrm{Hom}^{\mathrm{Cat}}_{\mathcal{C}}(X,Y)$$

There are some immediate notations.

Notation 2.9. Let Cat_2 be defined as above.

(a) There is an underlying functor $\underline{(-)}\colon \mathrm{Cat}_2 \to \mathrm{Cat}$ inherited from the enriched structure. The underlying category $\underline{\mathcal{C}}$ has the same object as that of $\mathcal{C}$, with $\mathrm{Map}_{\underline{\mathcal{C}}}(X,Y) \simeq \mathrm{Fun}_{\mathcal{C}}(X,Y)^{\simeq}$.

(b) A 2-functor $F\colon \mathcal{C}_1 \to \mathcal{C}_2$ is called fully faithful if for every $X, Y \in \mathcal{C}_1$, the induced functor

$$\mathrm{Fun}_{\mathcal{C}_1}(X,Y) \to \mathrm{Fun}_{\mathcal{C}_2}(X,Y)$$

is an equivalence. F is called essentially surjective if its underlying functor is essentially surjective. We say that F is an equivalence if it is fully faithful and essentially surjective.

(c) Let $\mathcal{C}$ be a 2-category. We denote by $\mathcal{C}^{\mathrm{op}}$ the 1-opposite of $\mathcal{C}$, which has the same object as $\mathcal{C}$, while the morphism category is given by

$$\mathrm{Fun}_{\mathcal{C}^{\mathrm{op}}}(X,Y) \simeq \mathrm{Fun}_{\mathcal{C}}(Y,X)$$

(d) Let $\mathcal{C}$ be a 2-category. We define its 2-opposite to be

$$\mathcal{C}^{\mathrm{co}} := \tau_{(-)^{\mathrm{op}}}(\mathcal{C}) \in \mathrm{Cat}_2$$

It has the same object as $\mathcal{C}$, while on morphism categories we always have

$$\mathrm{Fun}_{\mathcal{C}^{\mathrm{co}}}(X,Y) \simeq \mathrm{Fun}_{\mathcal{C}}(X,Y)^{\mathrm{op}}$$

An important concept of 2-categories is the notation of adjunctions. As expected, this is simply a generalization of categorical adjunctions.

Definition 2.10. Let $\mathcal{C}$ be a 2-category and let $X, Y \in \mathcal{C}$. We say that a morphism $f\colon Y \to X$ is a *left adjoint* if there is some $g\colon Y \to X$, called the *right adjoint* of f, and 2-morphisms $\eta\colon \mathrm{id}_Y \to g \circ f$ and $\epsilon\colon f \circ g \to \mathrm{id}_X$, called *unit* and *counit* respectively, such that there are commutative diagrams

$$\begin{array}{ccc} f & \xrightarrow{f\eta} & fgf \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \epsilon f} \\ & & f \end{array} \qquad \begin{array}{ccc} g & \xrightarrow{\eta g} & gfg \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle g\epsilon} \\ & & g \end{array}$$

We denote by $\mathrm{Fun}^L_{\mathcal{C}}(Y,X)$ and $\mathrm{Fun}^R_{\mathcal{C}}(Y,X)$ the subcategory of $\mathrm{Fun}_{\mathcal{C}}(Y,X)$ that consists of left adjoint and right adjoint morphisms.

One can show that 2-functors naturally preserves adjunctions. It follows immediately from the definition that a functor between 1-categories is a left adjoint functor if and only if it becomes a left adjoint in 2-categorical sense. The relationship can be written more explicitly as the following lemma.

Lemma 2.11. *Let $f\colon Y \rightleftharpoons X\colon g$ be a pair of morphisms in a 2-category $\mathcal{C}$ and let $\eta\colon \mathrm{id}_X \to g \circ f$ be a 2-morphism. The followings are equivalent.*

(1) f is left adjoint to g with unit η.

(2) For every $Z \in \mathcal{C}$, composition with f and g induces an adjunction

$$f_*\colon \mathrm{Fun}_{\mathcal{C}}(Z,Y) \rightleftharpoons \mathrm{Fun}_{\mathcal{C}}(Z,X)\colon g_*$$

with unit induced by η.

(3) For every $Z \in \mathcal{C}$ and morphisms $h\colon Z \to Y$ and $h'\colon Z \to X$, the map

$$\mathrm{Map}(fh, h') \to \mathrm{Map}(h, gh')$$

induced by η js an equivalence of spaces.

Proof. One can check that the functor $\mathrm{Fun}(Z,-)\colon \mathcal{C} \to \mathrm{Cat}$ is a 2-functor. Therefore, it preserves adjunctions. Thus, (1) implies (2). (2) and (3) are equivalent simply as the classical adjunction formula. It remains to prove that (3) implies (1). Let $\epsilon\colon fg \to \mathrm{id}_X$ be a preimage of id_g under the natural isomorphism $\mathrm{Map}(fg, \mathrm{id}) \xrightarrow{\simeq} \mathrm{Map}(g,g)$. It can be checked that the tuple (f,g,η,ϵ) provides the necessary data for an adjunction between f and g. □

We summarize some properties of adjunctions as follows. We state them in left adjoint's viewpoint and the right adjoint is absolutely dual.

Proposition 2.12. *Let $\mathcal{C}$ be a 2-category.*

(a) A morphism $f \in \mathcal{C}$ is left adjoint if and only if it is right adjoint as a morphism in $\mathcal{C}^{\mathrm{co}}$.

(b) A morphism $f \in \mathcal{C}$ is left adjoint if and only if $f^{\mathrm{op}} \in \mathcal{C}^{\mathrm{op}}$ is right adjoint.

(c) The composition of two left adjoint morphisms is left adjoint.
(d) Every equivalence is both a left adjoint and a right adjoint.
(e) The left adjoint of a morphism is unique up to equivalence.
(f) For $X, Y \in \mathcal{C}$, there is a natural equivalence

$$\mathrm{Fun}^L_{\mathcal{C}}(Y, X) \simeq \mathrm{Fun}^R_{\mathcal{C}}(X, Y)^{\mathrm{op}}$$

We will use the following criteria for adjunctions.

Proposition 2.13 ([HM24]; Proposition D.2.8.)**.** *Let $f\colon Y \to X$ be a morphism in a 2-category $\mathcal{C}$, the followings are equivalent:*

(1) f is a left adjoint morphism.
(2) For $Z \in \{X, Y\}$ the right adjoint G_Z of $f_*\colon \mathrm{Fun}_{\mathcal{C}}(Z, Y) \to \mathrm{Fun}_{\mathcal{C}}(Z, X)$ exists and the natural map $G_X(\mathrm{id}_X) \circ f \to G_Y(f)$ in $\mathrm{Fun}_{\mathcal{C}}(Y, Y)$ becomes an equivalence after applying $\mathrm{Map}(\mathrm{id}_Y, -)$.
(3) For all $Z \in \mathcal{C}$ the right adjoint G_Z of $f_*\colon \mathrm{Fun}_{\mathcal{C}}(Z, Y) \to \mathrm{Fun}_{\mathcal{C}}(Z, X)$ exists and for all $h\colon Z \to Z'$ in $\mathcal{C}$ the diagram

$$\begin{array}{ccc} \mathrm{Fun}_{\mathcal{C}}(Z', Y) & \xleftarrow{G_{Z'}} & \mathrm{Fun}_{\mathcal{C}}(Z', X) \\ {\scriptstyle h^*}\downarrow & & \downarrow{\scriptstyle h^*} \\ \mathrm{Fun}_{\mathcal{C}}(Z, Y) & \xleftarrow[G_Z]{} & \mathrm{Fun}_{\mathcal{C}}(Z, X) \end{array}$$

commutes and the natural map $h^* G_{Z'} \to G_Z h^*$ is an equivalence.

In this case, the right adjoint of f is given by $G_X(\mathrm{id}_X)\colon X \to Y$.

Remark 2.14. We can also obtain a notation of symmetric monoidal 2-category, which is a 2-category equipped with an operation $\otimes$ and some compatibility. For example, we need such an operation to be natural, that means, for any $W \to Z \in \mathcal{C}$, we have the following square

$$\begin{array}{ccc} W & \xrightarrow{\sigma_W} & W \otimes Y \\ {\scriptstyle P}\downarrow & \swarrow_{\sigma_P} & \downarrow{\scriptstyle f\otimes Y} \\ Z & \xrightarrow[\sigma_Z]{} & Z \otimes Y \end{array}$$

which commutes up to a 2-equivalence σ_P

2.2. **The category of kernels.** We now introduce the notation of 2-category of kernels, which is our main object to study.

Notation 2.15. We say a geometric setup $(\mathcal{C}, \mathcal{C}_0)$ is *full* if $\mathcal{C}_0$ contains all edges of $\mathcal{C}$.

Note that for any geometric setup $(\mathcal{C}, \mathcal{C}_0)$ and an object $S \in \mathcal{C}$, $((\mathcal{C}_0)_{/S}, (\mathcal{C}_0)_{/S})$ is a full geometric setup.

We begin with the formal definitions.

Definition 2.16. Let $\mathrm{D}(-)$ be a 3-functor formalism over $(\mathcal{C}, \mathcal{C}_0)$.

(1) Suppose $(\mathcal{C}, \mathcal{C}_0)$ is full, then $\mathrm{Span}(\mathcal{C}, \mathcal{C}_0) \simeq \mathrm{Span}(\mathcal{C})$ is self-enriched by Corollary 1.14. Then by Definition 2.6, $\mathrm{D}(-)$ determines a transfer of enrichment

$$\tau_{\mathrm{D}}\colon \mathrm{Enr}_{\mathrm{Span}(\mathcal{C})} \to \mathrm{Enr}_{\mathrm{Cat}} = \mathrm{Cat}_2$$

We denote by $\mathcal{K}_{\mathrm{D}} := \tau_{\mathrm{D}}(\mathrm{Span}(\mathcal{C})) \in \mathrm{Cat}_2$ and call it the *2-category of kernels* associated to $\mathrm{D}(-)$.

(2) In general, for any $S \in \mathcal{C}$, the functor $((\mathcal{C}_0)_{/S}, (\mathcal{C}_0)_{/S}) \to (\mathcal{C}, \mathcal{C}_0)$ determines a lax symmetric monoidal functor $\mathrm{Span}((\mathcal{C}_0)_{/S}, (\mathcal{C}_0)_{/S}) \to \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$, hence a 3-functor formalism $\mathrm{D}_S(-)$ over $((\mathcal{C}_0)_{/S}, (\mathcal{C}_0)_{/S})$ induced by $\mathrm{D}(-)$. We denote by $\mathcal{K}_{\mathrm{D},S} := \tau_{\mathrm{D}_S}(\mathrm{Span}((\mathcal{C}_0)_{/S}) \in \mathrm{Cat}_2$ and call it the *2-category of kernels* associated to $\mathrm{D}(-)$ over S. We will often abbriviate

$$\mathrm{Fun}_S(X, Y) := \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(X, Y)$$

and omit D when it is clear from the context.

Let us describe $\mathcal{K}_{\mathrm{D},S}$ more explicitly. The objects of $\mathcal{K}_{\mathrm{D},S}$ are morphisms $X \to S$ in $\mathcal{C}_0$. Given two objects $X, Y \in \mathcal{K}_{\mathrm{D},S}$, the mapping category is given by

$$\mathrm{Fun}_S(X, Y) = \mathrm{D}(X \times_S Y)$$

This formula comes from the description of transfer of enrichment and the fact that we have $\underline{\mathrm{Hom}}(X, Y) \simeq Y \times_S X$ in $\mathrm{Span}((\mathcal{C}_0)_{/S})$. Finally, suppose we have $M \colon X \to Y$ and $N \colon Y \to Z$ in $\mathcal{K}_{\mathrm{D},S}$, the composition $M \circ N$ is given by

$$M \circ N = \pi_{13!}(\pi_{12}^* M \otimes \pi_{23}^* N)$$

where π_{ij} denotes the projection on $X \times_S Y \times_S Z$.

From the fact that $\mathrm{Span}((\mathcal{C}_0)_{/S})^{\mathrm{op}} \simeq \mathrm{Span}((\mathcal{C}_0)_{/S})$ and that transfer of enrichment commutes with taking the opposite category, we have

Proposition 2.17. *Let $\mathrm{D}(-)$ be a 3-functor formalism on $(\mathcal{C}, \mathcal{C}_0)$, then for every $S \in \mathcal{C}$, we have*

$$\mathcal{K}_{\mathrm{D},S} \simeq \mathcal{K}_{\mathrm{D},S}^{\mathrm{op}}$$

Proposition 2.18. *Let $\mathrm{D}(-)$ be a 3-functor formalism on $(\mathcal{C}, \mathcal{C}_0)$. For every fixed $S \in \mathcal{C}$, we have a natural decomposition of $\mathrm{D}(-)$:*

$$\mathrm{Span}((\mathcal{C}_0)_{/S}) \xrightarrow{\Phi_{\mathrm{D},S}} \mathcal{K}_{\mathrm{D},S} \xrightarrow{\Psi_{\mathrm{D},S}} \mathrm{Cat}, \qquad \mathrm{D}(-) \text{ (over)}$$

where $\Phi_{\mathrm{D},S}$ is a 1-functor which can be natrually regarded as a 2-functor and $\Psi_{\mathrm{D},S}$ is a 2-functor between 2-categories.

Proof. This is by Lemma 2.7, where we can take $\alpha = \mathrm{D}(-)$, $F_\alpha = \Phi_{\mathrm{D},S}$ and $G_\alpha = \Psi_{\mathrm{D},S}$. □

More explicitly, $\Phi_{\mathrm{D},S}$ acts as identity on objects and sends $f = [X \leftarrow Y \to Z]$ to

$$\Phi_{\mathrm{D},S}(f) = f'_!(\mathbb{1}) \in \mathrm{D}(X \times_S Y) = \mathrm{Fun}_S(X, Y)$$

. The functor $\Psi_{\mathrm{D},S}$ is identified with $\mathrm{Fun}_S(S, -)$.

Our next aim is to show that the construction of the category of kernels $\mathcal{K}_{\mathrm{D},S}$ is functorial with respect to S, to $\mathrm{D}(-)$ and to the geometric setup $(\mathcal{C}, \mathcal{C}_0)$. We begin with the last two variants.

Proposition 2.19. *Let $\alpha \colon \mathrm{D}_1 \to \mathrm{D}_2$ be a natural transformation of 3-functor formalism on $(\mathcal{C}, \mathcal{C}_0)$. Then there is an induced 2-functor*

$$\varphi_\alpha \colon \mathcal{K}_{\mathrm{D}_1,S} \to \mathcal{K}_{\mathrm{D}_2,S}$$

for each $S \in \mathcal{C}$.

Proof. This follows immediately from the 2-functoriality of transfer of enrichment. □

Proposition 2.20. *Let $f : (\mathcal{C}, \mathcal{C}_0) \to (\mathcal{D}, \mathcal{D}_0)$ be a morphism of geometric setups and $\mathrm{D}(-)$ be a 3-functor formalism on $(\mathcal{D}, \mathcal{D}_0)$. Then precompose with f gives a 3-functor formalism $f^*\mathrm{D}(-)$ on $(\mathcal{C}, \mathcal{C}_0)$. Then f induces a 2-functor*

$$\rho_f \colon \mathcal{K}_{f^*\mathrm{D},S} \to \mathcal{K}_{\mathrm{D},f(S)}$$

for each $S \in \mathcal{C}$.

Proof. We consider the lax symmetric monoidal functor

$$\alpha_f \colon \mathrm{Span}((\mathcal{C}_0)_{/S}) \to \mathrm{Span}((\mathcal{D}_0)_{/S})$$

induced by f. By Lemma 2.7, there is a natural $\mathrm{Span}((\mathcal{D}_0)_{/S})$-enriched functor

$$G_{\alpha_f} \colon \tau_{\alpha_f}(\mathrm{Span}((\mathcal{C}_0)_{/S})) \to \mathrm{Span}((\mathcal{D}_0)_{/S})$$

After applying τ_{D}, we obtain the desired 2-functor ρ_f. □

We have the following concrete descriptions of φ_α and ρ_f. We can read off that φ_α acts on objects as identity and acts on morphism categories by

$$\mathrm{D}_1(X \times_S Y) \xrightarrow{\alpha} \mathrm{D}_2(X \times_S Y)$$

Moreover, ρ_f acts on objects by f and on morphism categories by

$$\mathrm{D}(f(X \times_S Y)) \xrightarrow{f_{XY!}} \mathrm{D}(f(X) \times_{f(S)} f(Y))$$

where $f_{XY} \colon f(X \times_S Y) \to f(X) \times_{f(S)} f(Y)$ is the natural map.

Remark 2.21. From the above explicit description, one can conclude that ρ_f and φ_α are compatible. That means, suppose that we have a natural transformation $\alpha \colon \mathrm{D}_1 \to \mathrm{D}_2$ of 3-functor formalism on $(\mathcal{D}, \mathcal{D}_0)$ and let $f : (\mathcal{C}, \mathcal{C}_0) \to (\mathcal{D}, \mathcal{D}_0)$ be a morphism of geometric setup, we have a commutative square of 2-functors:

$$\begin{array}{ccc} \mathcal{K}_{f^*\mathrm{D}_1,S} & \xrightarrow{\rho_f} & \mathcal{K}_{\mathrm{D}_1,f(S)} \\ {\scriptstyle \varphi_{f^*\alpha}}\downarrow & & \downarrow{\scriptstyle \varphi_\alpha} \\ \mathcal{K}_{f^*\mathrm{D}_2,S} & \xrightarrow[\rho_f]{} & \mathcal{K}_{\mathrm{D}_2,f(S)} \end{array}$$

The next aim is to show that this construction is functorial with respect to S. In order to construct the desired functor, we first need a lemma.

Lemma 2.22 (Proposition 2.2.14; [HM24])**.** *Let $F \colon (\mathcal{E}, \mathcal{E}_0) \to (\mathcal{C}, \mathcal{C}_0)$ be a morphism of geometric setups. Suppose additionally:*

(a) The functor $F \colon \mathcal{E} \to \mathcal{C}$ is a cartesian fibration;
(b) F admits coCartesian lifts of edges in $\mathcal{C}_0$ that lie in $\mathcal{E}_0$, i.e. for each $X \in \mathcal{E}$ and each edge $f \colon F(X) \to Y$ in $\mathcal{C}_0$, there exists an F-coCartesian edge $\overline{f} \colon X \to Y$ in $\mathcal{E}_0$ with $F(\overline{f}) = f$;
(c) Let $\sigma \colon [1] \times [1] \to \mathcal{E}$ be a commutative square such that the restrictions $\sigma|_{\{i\}\times[1]}$ are F-coCartesian edges of $\mathcal{E}$. Suppose that additionally $F \circ \sigma$ is a pullback square in $\mathcal{C}$. Then σ is a pullback square.

Then the induced functor $\mathrm{Span}(\mathcal{E}, \mathcal{E}_0) \to \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ is a coCartesian fibration.

Using this, we can first prove the category-level functoriality.

Lemma 2.23. *Let $(\mathcal{C},\mathcal{C}_0)$ be a geometric setup. Then there is a functor*

$$\mathrm{Span}(\mathcal{C},\mathcal{C}_0)\to \mathrm{Enr}_{\mathrm{Span}(\mathcal{C},\mathcal{C}_0)},\qquad S\mapsto \mathrm{Span}((\mathcal{C}_0)_{/S})$$

Sketch of proof. We recall the definition of the coCartesian operad in from [Lur17]. We have the following notations:

$$\mathrm{Fun}_{\mathrm{Fin}_*}(\mathcal{C}^{\mathrm{op}}\times \mathrm{Fin}_*,\mathcal{C}^{\mathrm{op},\sqcup})=\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}\times\Gamma^*,\mathcal{C}^{\mathrm{op}}),$$

Here Γ^* denotes the category with objects $(\langle n\rangle,i)$ for $\langle n\rangle\in\mathrm{Fin}_*$ and $i\in\{1,\dots,n\}$. The projection $\mathcal{C}^{\mathrm{op}}\times\Gamma^*\to\mathcal{C}^{\mathrm{op}}$ induces a functor $\mathcal{C}^{\mathrm{op}}\times\mathrm{Fin}_*\to\mathcal{C}^{\mathrm{op},\sqcup}$ over Fin_*. Let $\mathcal{E}'\subseteq\mathrm{Fun}([1],\mathcal{C})$ be the full subcategory spanned by the edges in $\mathcal{C}_0$ and we define

$$\mathcal{E}:=\left(\mathcal{E}'^{\mathrm{op},\sqcup}\times_{\mathcal{C}^{\mathrm{op},\sqcup}}(\mathcal{C}^{\mathrm{op}}\times\mathrm{Fin}_*)\right)^{\mathrm{op}}$$

which is induced by the target map. By construction there is a natural functor

$$t\colon\mathcal{E}\to\mathcal{C}\times\mathrm{Fin}_*^{\mathrm{op}}$$

Let $E_{\mathcal{E}}$ be the collection of edges in $\mathcal{C}_0$ of the form

$$\begin{array}{ccc}\oplus_{i=1}^m X_i & \longrightarrow & \oplus_{i=1}^m Y_i\\ \downarrow & & \downarrow\\ T & \longrightarrow & S\end{array}$$

with $T\to S\in\mathcal{C}_0$. It can be shown that $(\mathcal{E},E_{\mathcal{E}})$ is a geometric setup and t extends to a morphism of geometric setups

$$t\colon(\mathcal{E},E_{\mathcal{E}})\to(\mathcal{C},\mathcal{C}_0)\times(\mathrm{Fin}_*^{\mathrm{op}},\simeq).$$

By applying Span, we have

$$p\colon\mathcal{V}^{\otimes}:=\mathrm{Span}(\mathcal{E},E_{\mathcal{E}})\to\mathrm{Span}(\mathcal{C},\mathcal{C}_0)\times\mathrm{Fin}_*$$

Using Lemma 2.22, we can show that p is a $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$-coCartesian family of operads. Therefore, by straightening p we obtain a functor

$$F\colon\mathrm{Span}(\mathcal{C},\mathcal{C}_0)\to\mathrm{Op},\qquad S\mapsto\mathrm{Span}((\mathcal{C}_0)_{/S})^{\otimes}$$

One can base change from Fin_* to $\mathrm{Ass}^{\otimes}$ and show that F essentially factors through

$$G\colon\mathrm{Span}(\mathcal{C},\mathcal{C}_0)\to\mathrm{Enr}_{\mathrm{Span}(\mathcal{C},\mathcal{C}_0)},\qquad S\mapsto\mathrm{Span}((\mathcal{C}_0)_{/S})^{\otimes}$$

□

Replacing $(\mathcal{C},\mathcal{C}_0)$ by $(\mathcal{C}^{\mathrm{op},\sqcup,\mathrm{op}},\mathcal{C}_0)$, we can upgrade to a lax symmetric monoidal functor. Composing G with τ_{D}, we have:

Corollary 2.24. *Let $\mathrm{D}(-)$ be a 3-functor formalism over $(\mathcal{C},\mathcal{C}_0)$. Then the assignment $S\mapsto\mathcal{K}_{\mathrm{D},S}$ upgrades to a lax symmetric monoidal functor*

$$\mathcal{K}_{\mathrm{D},-}\colon\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}\to\mathrm{Cat}_2^{\times}$$

We have a more explicit description of the functor $\mathcal{K}_{\mathrm{D},-}$. For every morphism $S\xleftarrow{f}T\to T$, the induced 2-functor $f^*\colon\mathcal{K}_{\mathrm{D},S}\to\mathcal{K}_{\mathrm{D},T}$ sends X to $X\times_S T$ and the induced functor on morphism categories is the obvious induced map:

$$\mathrm{D}(X\times_S Y)\to\mathrm{D}\left((X\times_S T)\times_T(Y\times_S T)\right)$$

On the other side, for every morphism $T \leftarrow T \xrightarrow{g} S$, the induced functor acts as the forgetful functor and the induced functro on morphism categories is the obvious induced map:

$$\mathrm{D}(X \times_T Y) \to \mathrm{D}(X \times_S Y)$$

Remark 2.25. In Proposition 2.18, we constructed factorization functors $\Phi_{\mathrm{D},S}$ and $\Psi_{\mathrm{D},S}$. It can be shown that at least in some sense these two functors are functorial. More precisely, one can show that $\Phi_{\mathrm{D},-}$ becomes a morphism of lax symmetric monoidal functors from $\mathrm{Span}((\mathcal{C}_0)_{/(-)})$ to $\mathcal{K}_{\mathrm{D},-}$ between $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)^{\otimes}$ and $\mathrm{Cat}_2^{\times}$. Moreover, one can also show that any map $[T \xleftarrow{g} T' \xrightarrow{f} S]$ in $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$ induces a natural transformation of functors

$$\Psi_{\mathrm{D},T} \to \Psi_{\mathrm{D},S} \circ f_! g^*$$

Remark 2.26. By the explicit description of mapping categories of $\mathcal{K}_{\mathrm{D},T}$, one can show that for any $f\colon T \to S \in \mathcal{C}_0$ we have induced equivalences

$$\mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(f^*X, Y) \simeq \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},T}}(X, f_!Y), \quad \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(X, f^*Y) \simeq \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},T}}(f_!X, Y)$$

That is, the associated 2-functor f^* is left and right adjoint to $f_!$.

2.3. **Suaveness and Primness.** The adjoint properties in 2-categories make the category of kernels a fruitful notation. We now introduce the suave and prim objects and explain their properties.

Let X be an object in $\mathrm{Span}(\mathcal{C},\mathcal{C}_0)$. We have

$$\mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(X, S) \simeq \mathrm{D}(S \times_S X) \simeq \mathrm{D}(X) \simeq \mathrm{D}(X \times_S S) \simeq \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(S, X)$$

Definition 2.27. Let $\mathrm{D}(-)$ be a 3-functor formalism on $(\mathcal{C},\mathcal{C}_0)$. Let $f\colon X \to S$ be a morphism in $\mathcal{C}_0$ and fix $P \in \mathrm{D}(X)$.

(1) We say that P is *f-suave* if $P \in \mathrm{D}(X) \simeq \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(X, S)$ is a left adjoint from f to id. In this case, we denote its right adjoint by $\mathrm{SD}_f(P) \in \mathrm{D}(X)$ and call it the *f-suave dual* of P.
(2) Dually, we say that P is *f-prim* if $P \in \mathrm{D}(X) \simeq \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(X, S)$ is a right adjoint from f to id. In this case, we denote its left adjoint by $\mathrm{PD}_f(P) \in \mathrm{D}(X)$ and call it the *f-prim dual* of P.

Example 2.28. Let $X = S$ and $f = \mathrm{id}$. Then by definition $P \in \mathrm{D}(S)$ is f-suave if and only if it is f-prim if and only if it is dualizable.

It is easy to see that $\mathrm{SD}_f(-)$ and $\mathrm{PD}_f(-)$ become self-inverse functors when restrict to suave or prim objects. More precisely, if we use $\mathrm{SV}_f(X)$ and $\mathrm{PM}_f(X)$ to denote the full subcategories of $\mathrm{D}(X)$ consisting of the suave and prim objects, we have self inverse equivalences:

$$\mathrm{SD}(-)\colon \mathrm{SV}_f(X)^{\mathrm{op}} \xrightarrow{\simeq} \mathrm{SV}_f(X), \quad \mathrm{PD}(-)\colon \mathrm{PM}_f(X)^{\mathrm{op}} \xrightarrow{\simeq} \mathrm{PM}_f(X)$$

The following lemma shows that, an object is f-suave if and only if it satisfies specific projection formula with respect to f.

Lemma 2.29. *Let $\mathrm{D}(-)$ be a 6-functor formalism over $(\mathcal{C},\mathcal{C}_0)$. Suppose that $f\colon X \to S$ is a map in $\mathcal{C}_0$ and $P \in \mathrm{D}(X)$. Then P is f-suave if and only if the natural map*

$$\pi_1^* \underline{\mathrm{Hom}}(P, f^!\mathbb{1}) \otimes \pi_2^* P \to \underline{\mathrm{Hom}}(\pi_1^* P, \pi_2^! P)$$

becomes an isomorphism after applying $\underline{\mathrm{Hom}}(\mathbb{1},\Delta^!(-))$. *In this case, we have*

$$\mathrm{SD}_f(P)\simeq\underline{\mathrm{Hom}}(P,f^!\mathbb{1})$$

Moreover, for every $Z\in(\mathcal{C}_0)_{/S}$ *and* $M\in\mathrm{D}(Z)$, *we have equivalences*

$$\pi_1^*\underline{\mathrm{Hom}}(P,f^!\mathbb{1})\otimes\pi_2^*M\to\underline{\mathrm{Hom}}(\pi_1^*P,\pi_2^!M)$$

Proof. We can apply Proposition 2.13 to the 2-category $\mathcal{K}_{\mathrm{D},S}$ and $P\colon X\to S$. Note that the functor

$$P_*\colon\mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(Z,X)=\mathrm{D}(X\times_S Z)\to\mathrm{D}(Z)=\mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(Z,S)$$

is given by $\pi_{2!}(\pi_1^*P\otimes-)$, which must have a right adjoint. Therefore, P is a left adjoint if and only if we have $G_S(\mathrm{id}_S)\circ P\simeq G_X(P)$ in the setting of Proposition 2.13, which is exactly the morphism in the claim. □

Dually, we have:

Lemma 2.30. *Let* $\mathrm{D}(-)$ *be a 6-functor formalism over* $(\mathcal{C},\mathcal{C}_0)$. *Suppose that* $f\colon X\to S$ *is a map in* $\mathcal{C}_0$ *and* $P\in\mathrm{D}(X)$. *Then* P *is* f*-prim if and only if the natural map*

$$f_!(\pi_{2*}\underline{\mathrm{Hom}}(\pi_1^*P,\Delta_!\mathbb{1})\otimes P)\to f_*\underline{\mathrm{Hom}}(P,P)$$

becomes an isomorphism after applying $\underline{\mathrm{Hom}}(\mathbb{1},-)$. *Here* π_i *denotes the projections on* $X\times_S X$ *and* $\Delta:X\to X\times_S X$ *is the diagonal. In this case, we have*

$$\mathrm{PD}_f(P)\simeq\pi_{2*}\underline{\mathrm{Hom}}(\pi_1^*P,\Delta_!\mathbb{1}))$$

Moreover, for every $Z\in(\mathcal{C}_0)_{/S}$ *and* $M\in\mathrm{D}(Z)$, *we have equivalences*

$$\mathrm{pr}_2^!(\mathrm{pr}_1^*\pi_{2*}\underline{\mathrm{Hom}}(\pi_1^*P,\Delta_!\mathbb{1})\otimes M)\to\mathrm{pr}_{2*}\underline{\mathrm{Hom}}(\mathrm{pr}_1^*P,M)$$

where pr_1 *and* pr_2 *denote two projections on* $X\times_S Z$.

Corollary 2.31. *Let* $\mathrm{D}(-)$ *be a 6-functor formalism over* $(\mathcal{C},\mathcal{C}_0)$. *Suppose that* $f\colon X\to S$ *is a map in* $\mathcal{C}_0$ *and* $P\in\mathrm{D}(X)$. *Then*

(1) If P *is* f*-suave then for all* $M\in\mathrm{D}(S)$, *the natural map*

$$\mathrm{SD}_f(P)\otimes f^*M\xrightarrow{\simeq}\underline{\mathrm{Hom}}(P,f^!M)$$

is an equivalence.

(2) If P *is* f*-prim then for all* $M\in\mathrm{D}(X)$, *the natural map*

$$f_!(\mathrm{PD}_f(P)\otimes M)\xrightarrow{\simeq}f_*\underline{\mathrm{Hom}}(P,M)$$

is an equivalence.

The suave objects can also interact with prim objects.

Lemma 2.32. *Let* $\mathrm{D}(-)$ *be a 6-functor formalism over* $(\mathcal{C},\mathcal{C}_0)$. *Let* $f\colon X\to S$ *be a map in* $\mathcal{C}_0$, $P\in\mathrm{D}(X)$ *be* f*-suave and* $Q\in\mathrm{D}(X)$ *be* f*-prim. Then we have a natural equivalence*

$$f_*\underline{\mathrm{Hom}}(Q,\mathrm{SD}_f(P))\simeq f_*\underline{\mathrm{Hom}}(\mathrm{PD}_f(Q),P)^\vee$$

Proof. By Corollary 2.31, we have

$$\begin{aligned}f_*\underline{\mathrm{Hom}}(Q,\mathrm{SD}_f(P))\simeq&f_!(\mathrm{PD}_f(Q)\otimes\mathrm{SD}_f(P))\simeq\mathrm{PD}_f(Q)\circ\mathrm{SD}_f(P)\\\simeq&(Q\circ P)^\vee\simeq(f_!(Q\otimes P))^\vee\simeq f_*\underline{\mathrm{Hom}}(\mathrm{PD}_f(Q),P)^\vee\end{aligned}$$

□

We now explain the suave and prim maps.

Definition 2.33. Let $\mathrm{D}(-)$ be a 3-functor formalism on a geometric setup $(\mathcal{C}, \mathcal{C}_0)$. Let $f\colon Y \to X$ be a morphism in $\mathcal{C}_0$.

(1) We say that f is D-suave if $\mathbb{1} \in \mathrm{D}(Y)$ is f-suave. In this case we will denote by $\omega_f := \mathrm{SD}_f(\mathbb{1}) \in \mathrm{D}(Y)$ and call it the *dualizing complex* of f. We say that f is D-smooth if it is D-suave and ω_f is invertible.
(2) Dually, we say that f is D-prim if $\mathbb{1} \in \mathrm{D}(Y)$ is f-prim. In this case we will denote by $\delta_f := \mathrm{SD}_f(\mathbb{1}) \in \mathrm{D}(Y)$ and call it the *codualizing complex* of f.

We also have these criteria for suave and prim maps.

Lemma 2.34. *Let $\mathrm{D}(-)$ be a 6-functor formalism on a geometric setup $(\mathcal{C}, \mathcal{C}_0)$. Let $f\colon Y \to X$ be a morphism in $\mathcal{C}_0$. Then f is D-suave if and only if we have a natural equivalence $\pi_1^* f^! \mathbb{1} \simeq \pi_2^! \mathbb{1}$, where π_i denote the projections on $Y \times_X Y$. In this case, we have*

$$\omega_f \simeq f^! \mathbb{1}$$

Proof. This follows by applying $P = \mathbb{1}$ to Lemma 2.29. □

Dually, we have:

Lemma 2.35. *Let $\mathrm{D}(-)$ be a 6-functor formalism on a geometric setup $(\mathcal{C}, \mathcal{C}_0)$. Let $f\colon Y \to X$ be a morphism in $\mathcal{C}_0$. Then f is D-prim if and only if we have a natural equivalence $f_! \pi_{2*} \Delta_! \mathbb{1} \simeq f_* \mathbb{1}$, where π_i denote the projections on $Y \times_X Y$ and Δ is the diagonal map on $Y \times_X Y$. In this case, we have*

$$\delta_f \simeq \pi_{2*} \Delta_! \mathbb{1}$$

Therefore, one can deduce from above criteria or directly from definitions that suave and prim maps are stable under base change and composition.

Corollary 2.36. *Let $\mathrm{D}(-)$ be a 6-functor formalism over $(\mathcal{C}, \mathcal{C}_0)$. Let $f\colon Y \to X$ be a morphism in $\mathcal{C}_0$. Then*

(1) If f is D-suave then we have equivalences of functors $\mathrm{D}(X) \to \mathrm{D}(Y)$:

$$\omega_f \otimes f^* \simeq f^!, \quad f^* \simeq \underline{\mathrm{Hom}}(\omega_f, f^!)$$

(2) If f is D-prim then we have equivalences of functors $\mathrm{D}(Y) \to \mathrm{D}(X)$:

$$f_!(\delta_f \otimes -) \simeq f_*, \quad f_! \simeq f_* \underline{\mathrm{Hom}}(\delta_f, -)$$

Proof. This follows by applying Corollary 2.31 to $P = \mathbb{1}$. □

An important property of suave and prim maps is that they allow base change formulas in more general forms. By applying the unit and counit maps in a 6-functor formalism, one can obtain a list of base change natural transformations. Suppose that we assert some suave or prim conditions, then we may obtain better-behaved base-change formulas.

Proposition 2.37. *Let $\mathrm{D}(-)$ be a 6-functor formalism over a geometric setup $(\mathcal{C}, \mathcal{C}_0)$. Suppose we have a Cartesian square in $\mathcal{C}_0$:*

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ X' & \xrightarrow[g]{} & X \end{array}$$

(1) *Assume that g is* D*-suave, then the following natural transformations*
$$g^*f_* \to f'_*g'^*, \quad f'_!g'^! \to g^!f_!, \quad f'^*g^! \to g'^!f^*, \quad g'^*f^! \to f'^!g^*$$
are equivalences.

(2) *Assume that g is* D*-prim, then the following natural transformations*
$$g'_!f'^! \to f^!g_!, \quad f^*g_* \to g'_*f'^*, \quad f_!g'_* \to g_*f'_!, \quad g'_!f'_* \to f_*g'_!$$
are equivalences.

Proof. We show the first and the third equivalences in (1), the rests are by similar argument. By Corollary 2.36, $g^* \simeq \underline{\mathrm{Hom}}(\omega_g, g^!)$ admits a left adjoint $g_\natural = g_!(-\otimes\omega_g)$. Since suaveness is stable under base change, g' is also suave and admits a left adjoint $g'_\natural$. Therefore, it suffices to show that the natural map $g'_\natural f'^* \to f^*g_\natural$ is an equivalence. Then apply ??? to the adjunction pair
$$\omega_g \colon X' \rightleftharpoons X : \mathbb{1}_{X'}$$
in 2-category $\mathcal{K}_{\mathrm{D},X}$ and after applying $\Psi_{\mathrm{D},X} \colon \mathcal{K}_{\mathrm{D},X} \to \mathrm{Cat}$, we obtain the desired equivalence
$$g'_\natural f'^* \xrightarrow{\simeq} f^*g_\natural$$
Similarly, we can apply Remark 2.14 to
$$\mathbb{1}_{X'} \colon X' \rightleftharpoons X : \omega_g$$
which yields an equivalence $f'^*g^! \xrightarrow{\simeq} g'^!f^*$ as desired.

The other claims are obtained by applying the same procedure to $\mathcal{K}^{\mathrm{op}}_{\mathrm{D},X}$, $\mathcal{K}^{\mathrm{co}}_{\mathrm{D},X}$ and $\mathcal{K}^{\mathrm{co,op}}_{\mathrm{D},X}$. □

2.4. **étale and proper maps.** We now briefly introduce the definition of a special class of suave and prim maps, namely the étale and proper maps, respectively.

Definition 2.38. Let $\mathrm{D}(-)$ be a 3-functor formalism over $(\mathcal{C},\mathcal{C}_0)$ and let $f \colon Y \to X$ be a truncated map in $\mathcal{C}_0$. Let $\Delta_f \colon Y \to Y \times_X Y$ denote the diagonal map.

(1) We say that f is D-*étale* if it is D-suave and Δ_f is D-étale or equivalence.
(2) We say that f is D-*proper* if it is D-prim and Δ_f is D-proper or equivalence.

Note that f being n-truncated implies Δ_f being $n-1$-truncated. The definition is meaningful since we can repeat until we end up with an equivalence.

We states the properties of étale and proper maps.

Proposition 2.39. *Let* $\mathrm{D}(-)$ *be a 3-functor formalism over* $(\mathcal{C},\mathcal{C}_0)$.
(a) D*-étale maps are stable under composition and base change.*
(b) If $f \colon Y \to X$ is D*-étale, then $P \in \mathrm{D}(Y)$ is f-suave if and only if it is dualizable, in which case we have $\mathrm{SD}_f(P) \simeq P^\vee$. The similar is true for proper maps.*

The étaleness and properness can also help us to compare the "$*$" with "!". We have the following identifications:

Proposition 2.40. *Let* $\mathrm{D}(-)$ *be a 6-functor formalism over* $(\mathcal{C},\mathcal{C}_0)$ *and let $f \colon Y \to X$ be a map in $\mathcal{C}_0$. Suppose that Δ_f is* D*-étale, then there exists a natural map $f^! \to f^*$. Moreover, the followings are equivalent:*

(1) f is D*-étale.*
(2) $f^!\mathbb{1} \to f^*\mathbb{1}$ is an equivalence in $\mathrm{D}(X)$.

(3) The natural map $f^! \to f^*$ is an equivalence.

The similar is true for proper maps where we replace every f^* by f_* and every $f^!$ by $f_!$.

3. Descent properties

We briefly introduce the descent properties of 6-functor formalisms, and how these descent properties translate into 2-categorical information. Let $\mathcal{C}$ be a category and $\mathcal{U}$ be a sieve over $U \in \mathcal{C}$. Recall that a functor $\mathcal{F}\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{V}$ descends universally along $\mathcal{U}$ if it descends along every pullback of $\mathcal{U}$.

Proposition 3.1. *Let $\mathrm{D}(-)\colon \mathrm{Span}(\mathcal{C}, \mathcal{C}_0)$ be a 3-functor formalism with $\mathrm{D}^*\colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Cat}$ the restriction functor. Let $\mathcal{U}$ be a sieve over $X \in \mathcal{C}$ such that D^* descends universally along $\mathcal{U}$. Then the induced 2-functor*

$$\mathcal{K}_{\mathrm{D},X} \to \lim_{U \in \mathcal{U}^{\mathrm{op}}} \mathcal{K}_{\mathrm{D},U}$$

is fully faithful.

Proof. This is almost by definition. For $Y, Z \in \mathcal{K}_{\mathrm{D},X}$, the descent of D^* gives an equivalence

$$\mathrm{Fun}_{\mathcal{K}_{\mathrm{D},S}}(Z, Y) = \mathrm{D}(Y \times_X Z) \simeq \lim_{U \in \mathcal{U}^{\mathrm{op}}} \mathrm{D}((Y \times_X Z) \times U) = \lim_{U \in \mathcal{U}^{\mathrm{op}}} \mathrm{Fun}_{\mathcal{K}_{\mathrm{D},U}}(Z \times_X U, Y \times_X U)$$

which is exactly the fully faithfulness. □

Corollary 3.2. *Suppose that $\mathrm{D}(-)$ is a 3-functor formalism on $(\mathcal{C}, \mathcal{C})$. Let $\mathcal{U}$ be a sieve over $X \in \mathcal{C}$ such that D^* descends universally along $\mathcal{U}$. Then the natural map*

$$\operatorname*{colim}_{U \in \mathcal{U}^{\mathrm{op}}} U \xrightarrow{\simeq} X$$

is an equivalence in $\mathcal{K}_{\mathrm{D}}$.

Proof. It suffices to show that for every $Z \in \mathcal{K}_{\mathrm{D}}$, we have an equivalence

$$\lim_{U \in \mathcal{U}^{\mathrm{op}}} \mathrm{D}(Z \times U) \simeq \lim_{U \in \mathcal{U}^{\mathrm{op}}} \mathrm{Fun}_{\mathcal{K}_{\mathrm{D}}}(U, Z) \simeq \mathrm{Fun}_{\mathcal{K}_{\mathrm{D}}}(Z, X) \simeq \mathrm{D}(Z \times X)$$

which is exactly the descent information. □

For suave and prim objects, we also have descent properties.

Proposition 3.3. *Let $\mathrm{D}(-)$ be a 3-functor formalism over $(\mathcal{C}, \mathcal{C}_0)$. Let $f\colon X \to S$ be a morphism in $\mathcal{C}_0$ and let $P \in \mathrm{D}(\mathrm{X})$. For $g\colon S' \to S$, we denote $g'\colon X' := X \times_S S' \to X$ and $f'\colon X' \to S'$ the base change. Then we have:*

1. *If P is f-suave, then for every $g\colon S' \to S$, $g'^* P$ is f'-suave.*
2. *Let $\mathcal{U}$ be a sieve on S, along which D^* descends universally. If for every $g\colon U \to S$ in $\mathcal{U}$, $g'^* P$ is f'-suave, then P is f-suave.*

The same is true for primness.

Proof. The first claim follows from Corollary 2.24. The functor $g^*\colon \mathcal{K}_{\mathrm{D},S} \to \mathcal{K}_{\mathrm{D},S'}$ sending P to $g'^* P$ is naturally a 2-functor, thus preserves adjunction. The second claim follows from Proposition 3.1. □

The suave and prim objects can also be detected on the suave and prim covers.

Theorem 3.4. *Let D be a 3-functor formalism on a geometric setup $(\mathcal{C}, \mathcal{C}_0)$, let $f\colon X \to S$ be a map in $\mathcal{C}_0$, let $P \in \mathrm{D}(X)$ and let $\mathcal{U} \subset (\mathcal{C}_0)_{/X}$ be a sieve along which D^* descends universally.*

(1) *Suppose that* $\mathrm{D}(X)$ *and* $\mathrm{D}(S)$ *admit* $\mathcal{U}$*-indexed colimits and* $f_!(P\otimes -)\colon \mathrm{D}(X)\to \mathrm{D}(S)$ *preserves them. If* $\mathcal{U}$ *is generated by maps* $g\colon U\to X$ *such that* $\mathbb{1}$ *is* g*-suave and* g^*P *is* fg*-suave, then* P *is* f*-suave and*

$$\mathrm{SD}_f(P)=\operatorname*{colim}_{g\in\mathcal{U}} g_!\mathrm{SD}_{fg}(g^*P)$$

(2) *Suppose that* $\mathrm{D}(X)$ *and* $\mathrm{D}(S)$ *admit* $\mathcal{U}^{\mathrm{op}}$*-indexed limits and* $f_!(P\otimes -)\colon \mathrm{D}(X)\to \mathrm{D}(S)$ *preserves them. If* $\mathcal{U}$ *is generated by maps* $g\colon U\to X$ *such that* $\mathbb{1}$ *is* g*-prim and* g^*P *is* fg*-prim then* P *is* f*-prim and*

$$\mathrm{PD}_f(P)=\lim_{g\in\mathcal{U}^{\mathrm{op}}} g_!\mathrm{D}_{fg}(g^*P)$$

sketch of proof. Since the two claims are dual. It suffices to prove the first claim. We note that by Corollary 3.2, we have $X=\operatorname*{colim}_{U\in\mathcal{U}} U$ in $\mathcal{K}_{\mathrm{D},S}$. Pick a generating family $(g_i\colon U_i\to X)_{i\in I}$ of $\mathcal{U}$. Then by general facts about descent, we can write X as a colimit of an I-indexed simplicial diagram. By the above proposition, it now suffices to show that for every $g_{i_\bullet}\colon U_{i_\bullet}:=U_{i_1}\times_X\cdots\times_X U_{i_n}$, the object $g_{i_\bullet}^*P$ is $fg_{i_\bullet}$-suave. This can be proven through induction on n. □

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